

Georgia Institute of Technology



Experimental Investigation of Bending and Tensile Strength of 3D Printed Specimens

Varying Infill Geometries

AE 3610 Final Project

Abstract. This experiment investigates how infill geometry affects the bending and tensile mechanical performance of 3D printed PLA specimens. Grid, honeycomb, and triangular infill patterns were printed at 10% infill and tested through four-point bending and uniaxial tension. Digital Image Correlation and MATLAB data reduction were used to determine compressive and tensile elastic moduli, while print time and fracture behavior were compared across geometries.

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Abstract

The experiment objective was to investigate how infill geometry impacts bending and tensile mechanical performance of 3D printed PLA parts and to compare their elastic moduli and print times. Three geometries, grid, honeycomb, and triangular infills, were printed using a Bambu Lab X1E with 10% infill and tested using four-point bending and uniaxial tension tests. Digital Image Correlation (DIC) was used to measure strain over the test section. MATLAB data reduction was employed after removing low-quality regions caused by speckle delamination. Bending behavior was calculated using the axial strain distribution to calculate the compressive elastic modulus. Tensile modulus was determined from the average micro-strain and corresponding load cell data from the tensile test. Results show that honeycomb specimens produced the highest compressive modulus, while the triangular specimens produced the highest tensile modulus, and grid infill had the lowest stiffness values. Fracture loads varied significantly between identical test samples, most likely due to small print defects and failure at the interface between the clamp and specimen infill change. Overall, the results show that infill geometry plays a significant role in mechanical performance, with different patterns offering distinct advantages depending on the loading method.

Introduction

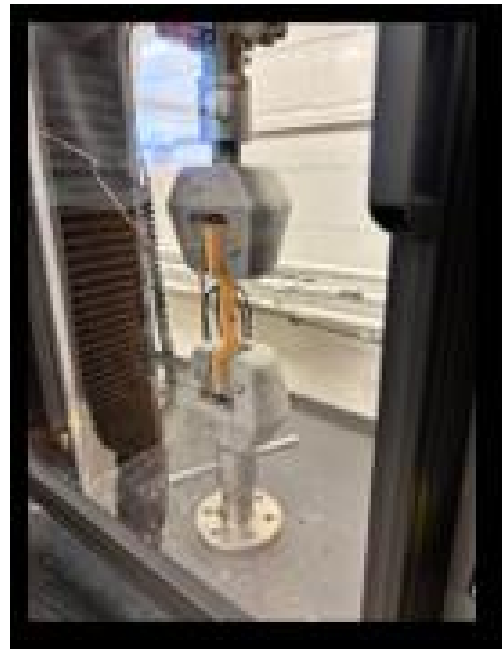
This experiment explored how different 3D printed designs behave when loaded in tension and bending. Dog bone specimens were pulled apart in a simple tension test, while beam specimens were tested using a four-point bending setup. All testing took place on the Instron machine in the Engineering Science and Mechanics Building at Georgia Tech. The specimens were 3D printed using three internal patterns: grid, honeycomb, and triangular. Each sample was placed into the machine and tested one at a time, and the force or moment on the specimen and displacement were recorded with the Instron. Strain was recorded by strain gauges and through analysis of adhered DIC patterns, both throughout multiple points of the elastic region, at the ultimate tensile strength or maximum force value recorded, and at the point of failure. By measuring how each specimen responded to these loads, the experiment aimed to compare the stiffness and overall performance of the three infill types.

Experimental Setup

All testing was conducted using the Instron load frame located in the Engineering Science and Mechanics Building on Georgia Tech's campus. Dog bone specimens were used for the tensile tests, and beam specimens were used for the four-point bending tests. The Instron machine provided controlled loading and displacement while a Digital Image Correlation camera system captured deformation throughout each experiment. The tension and bending setups are shown in Figure 2.1 to illustrate the orientation of each specimen during testing.



(a) Four-point bending beam test.



(b) Tensile dogbone test.

Figure 2.1: Experimental setup for four-point bending beam test and tensile dogbone test.

Measurements such as applied force for tension tests and applied moment for bending tests were collected at each image frame and at failure to ensure consistent comparisons among the different infill types. Complications arose during the manufacturing process; the prints were a little worse than hoped for, but a lot of copies were made and vetted to a certain quality. Some errors arose from testing as well. Initially, the fracture force of the specimens was unknown, and a couple of them were broken too quickly to collect data from them. This problem was also solved by printing extra copies.

All specimens were manufactured using a Bambu Lab X1E printer with standard PLA

filament. Each sample was printed with 10% infill using one of the three internal geometries examined in this study: grid, honeycomb, or triangular. The outer shell thickness, internal dimensions, and overall specimen geometry were kept consistent across all prints to isolate the effect of infill pattern. Physical measurements of each specimen were taken prior to testing and are provided in Tables 2.1 and 2.2.

Table 2.1: Bending Specimen Dimensions and Mass

Specimen	l (m)	w (m)	h (m)	Mass (kg)
Grid-1	0.2280	0.02527	0.02520	0.0394
Grid-2	0.2280	0.02527	0.02524	0.0389
Honeycomb-1	0.2280	0.02520	0.02530	0.045
Triangular-1	0.2275	0.02520	0.02526	0.0395

Table 2.2: Tensile Specimen Dimensions and Mass

Specimen	l (m)	w (m)	t (m)	Mass (kg)
Grid-1	0.07938	0.02489	0.00660	0.071
Grid-2	0.07938	0.02535	0.00650	0.071
Honeycomb-1	0.07779	0.02535	0.00650	0.071
Honeycomb-2	0.07925	0.02489	0.00650	0.071
Triangular-2	0.07938	0.02535	0.00650	0.070

During setup, attention was given to pin placement and loading alignment to ensure that each sample experienced the correct support and loading conditions. Overall, we observed consistent print quality across all twelve test specimens in terms of physical dimensions and masses.

We initially intended to use strain gauges as well to fully map out the stress-strain curve, but due to a miscommunication with the TA and duplicate final project proposals, we did not have strain gauges allocated despite submitting one project proposal requesting them. Thus, we were only allocated a few extra strain gauges, which we used to determine strain at certain points. Because we did not have sufficient gauges for thorough testing across all specimens, this analysis was left out of the Results and Discussion section, and we were forced to take DIC images at discrete epochs of loading instead of a complete mapping of stress-strain behavior.

Results and Discussion

The initial test involved two specimens for each of the tests: honeycomb, grid, and triangular infill for both the tension dog bone tests and bending beam tests. However, some samples were not considered due to poor data quality from DIC sticker delamination, primarily indicated by observed shadowy regions when image processing. Multiple DIC images were captured to ensure enough data was present; however, images at lower displacements were used for analysis since this resulted in less DIC pattern delamination while remaining safely in the elastic zone. Values recorded at fracture did not utilize DIC due to significant delamination and instead used data from the Instron's load cell.

The raw data was presented from the VIC-2D software as a matrix with a `.csv` extension containing x and y coordinates relative to the geometric centroid, x and y displacement values, and x and y strain values, where x is horizontal and y is vertical relative to each image. Some data contained zero values due to the manual removal of low-quality data zones that would incorrectly alter results, and this data was removed in MATLAB. MATLAB was used for all data analysis and figure creation. Additionally, the displacement values were removed since experimental strain is more important and already calculated.

3.1 Bending Beam Analysis

The bending beam data was analyzed first. The axial ϵ_{xx} microstrain was plotted as a contour map for the Grid-1, Grid-2, Honeycomb-1, and Triangular-1 samples, shown in Figures 3.1 through 3.4.

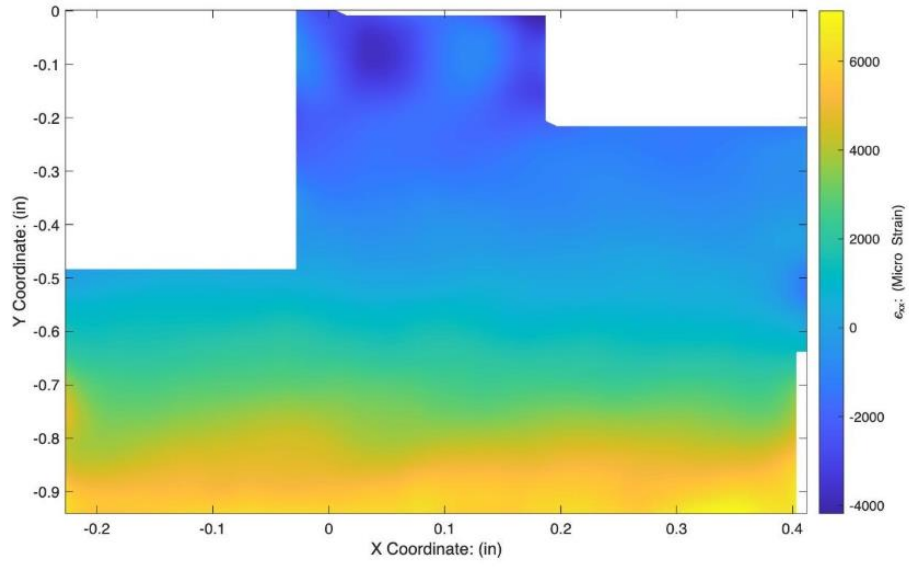


Figure 3.1: Grid-1 beam specimen axial microstrain ε_{xx} .

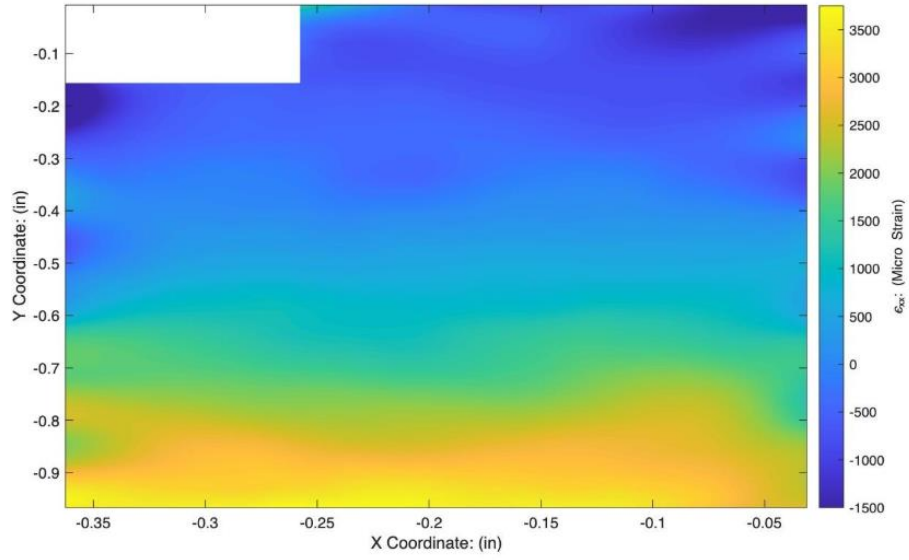


Figure 3.2: Grid-2 beam specimen axial microstrain ε_{xx} .

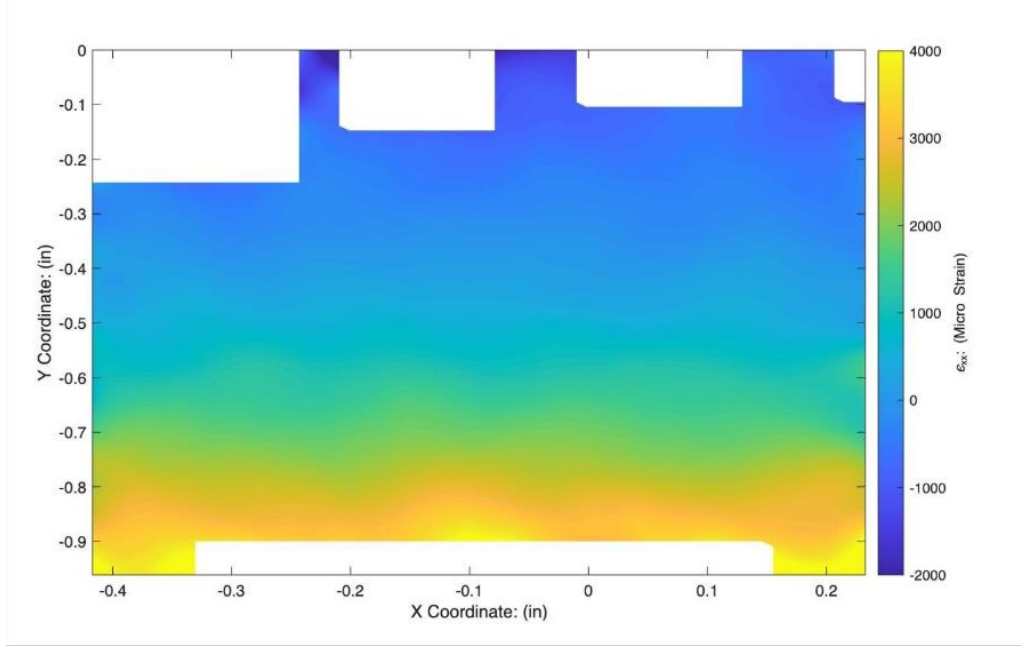


Figure 3.3: Honeycomb-1 beam specimen axial microstrain ϵ_{xx} .

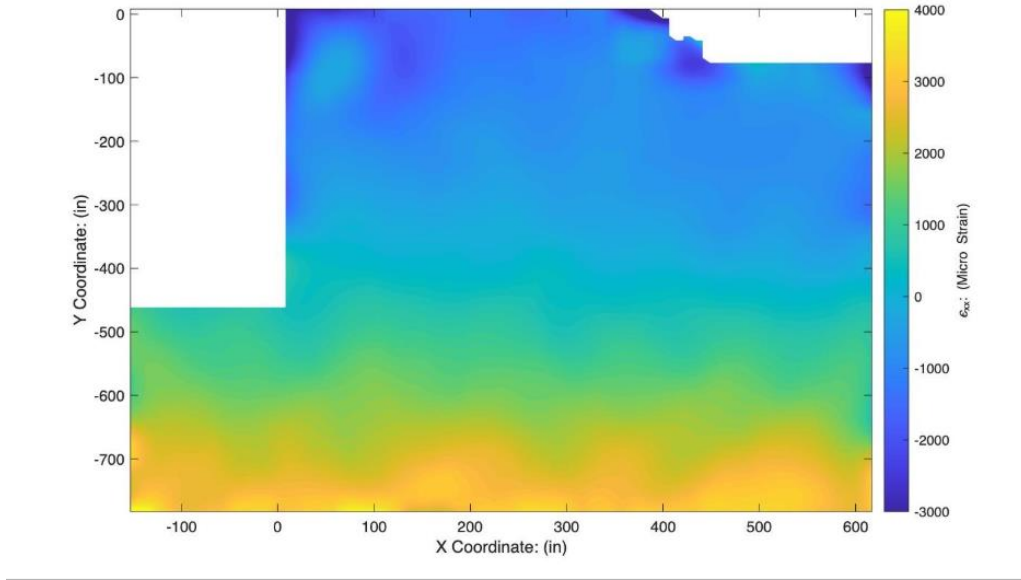


Figure 3.4: Triangular-1 beam specimen axial microstrain ϵ_{xx} .

Observed in all figures is that the lower surface is in a state of tension, indicated by positive microstrain, and the upper surface is in a state of compression, indicated by negative microstrain, with the region in between appearing approximately linear. This is predicted behavior for beams under an applied moment. Since the beams involved complicated geometry, they were assumed to be non-isotropic, and the axial strain equation in Equation 3.1

was used for analysis.

$$\varepsilon_{xx} = -\kappa y - b \quad (3.1)$$

The term κ is the bending curvature, and b is the strain offset. Three vertical lines at constant x-coordinate were selected to run across the height of the beam. These were selected at an approximately central location and at left-hand and right-hand locations relative to the center. This was designed to encapsulate an appropriate range of data in both the axial and transverse directions. These three data lines were concatenated and input into MATLAB's `polyfit()` and `polyval()` functions to output a linear equation representing the first-order line of best fit to the data. The data and line-of-best-fit results are shown as a function of the y-coordinate in Figures 3.5 through 3.8.

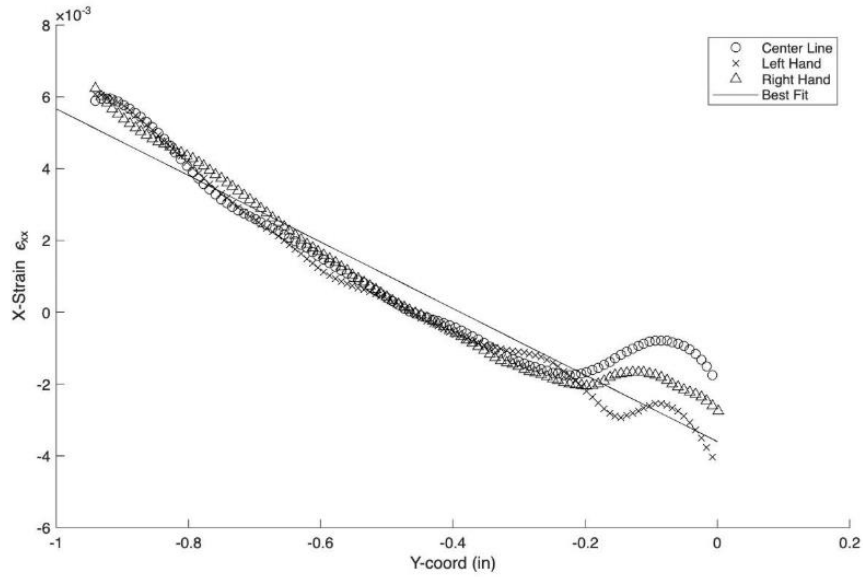


Figure 3.5: Bending specimen Grid-1 axial strain data as function of y-coordinate.

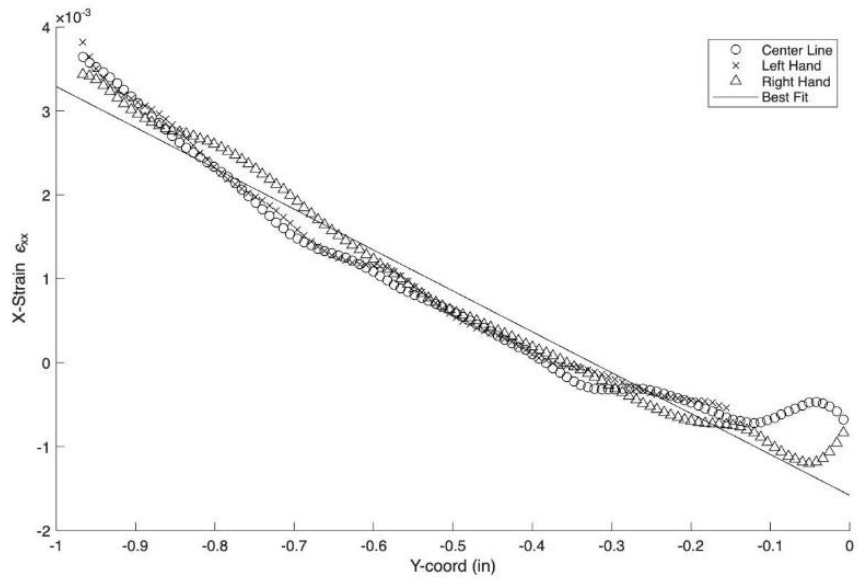


Figure 3.6: Bending specimen Grid-2 axial strain data as function of y-coordinate.

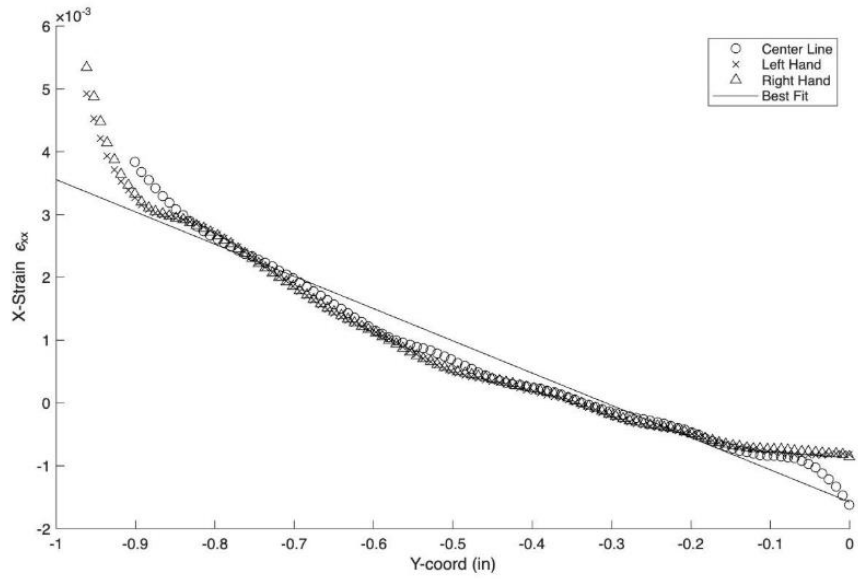


Figure 3.7: Bending specimen Honeycomb-1 axial strain data as function of y-coordinate.

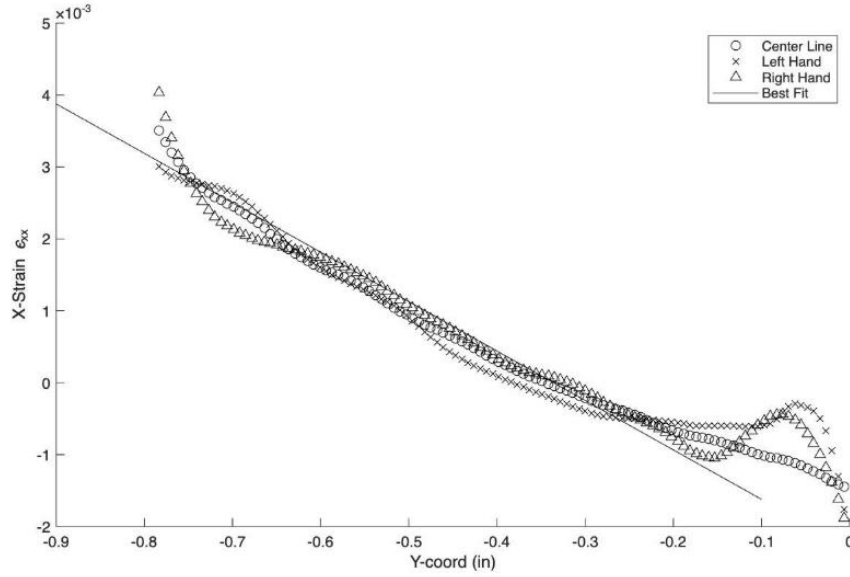


Figure 3.8: Bending specimen Triangular-1 axial strain data as function of y-coordinate.

The results for κ and b relating the output first-order linear equation to Equation 3.1 are found in Table 3.1.

Table 3.1: Bending Specimen Applied Moment Terms

Specimen	Second Moment of Area (m^4)	Curvature, κ ($1/\text{m}$)	Strain Offset, b	Applied Moment (Nm)
Grid-1	0.4890×10^{-9}	0.36492	0.0036038	77.794
Grid-2	0.4912×10^{-9}	0.19179	0.0015798	31.137
Honeycomb-1	0.4934×10^{-9}	0.20201	0.0015763	42.874
Triangular-1	0.4912×10^{-9}	0.21646	0.0016215	38.121

Next, the second moment of area for each beam was calculated using Equation 3.2.

$$I = \frac{wh^3}{12} \quad (3.2)$$

Here, h is the height and w is the width, which were approximately the same. The resultant second moments of area are found in Table 3.1. Since the moment was applied using a distributed force applied at a distance $a = 0.097$ m from each end of the specimen, this was also calculated for each test using Equation 3.3.

$$M = Fa \quad (3.3)$$

With the above calculations made from Equations 3.1, 3.2, and 3.3, and Table 2.1, the results were input into Equation 3.4.

$$E_C = \frac{M}{I\kappa} \left(1 - \frac{2b}{\kappa h}\right)^2 \quad (3.4)$$

The resulting compressive elastic modulus for each bending specimen is found in Table 3.2.

Table 3.2: Bending Specimen Calculated Strength and Build Properties

Specimen	E_C (MPa)	Moment at Fracture (Nm)	Print Time (mm:ss)
Grid-1	1.3700×10^5	122.414	47:40
Grid-2	1.2098×10^5	144.045	47:40
Honeycomb-1	1.6452×10^5	153.26	71:10
Triangular-1	1.4126×10^5	150.35	51:04

This also includes the provided print time for each infill geometry using Bambu Studio software on a Bambu Lab X1E 3D printer and the recorded applied moments at fracture using Equation 3.3. Fracture was recorded when fibers were first visibly separated.

3.2 Tensile Test Analysis

Next, the data from the tension tests were analyzed. It was determined that, due to the tendency for the DIC speckle pattern to delaminate from the part, it would be more useful to calculate the tensile elastic modulus through a direct tensile test rather than the bending test, which was helpful for finding the compressive elastic modulus. Since the tensile elastic modulus is calculated here, the strain in the y-direction, ε_{yy} , is considered. This microstrain value was plotted as a contour map for dog bone specimens Grid-1, Grid-2, Honeycomb-1, Honeycomb-2, and Triangular-2 in Figures 3.9 through 3.13.

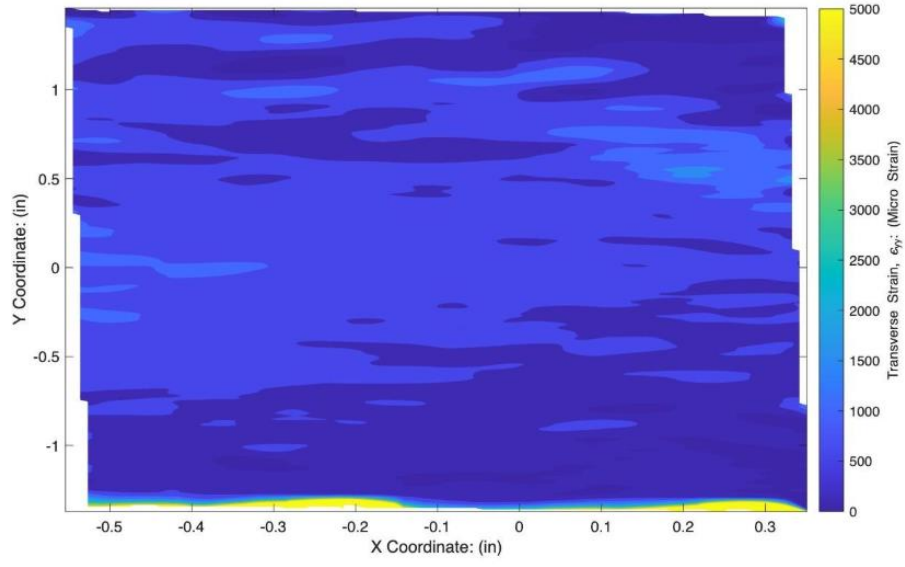


Figure 3.9: Tensile specimen Grid-1 ϵ_{yy} microstrain contour map.

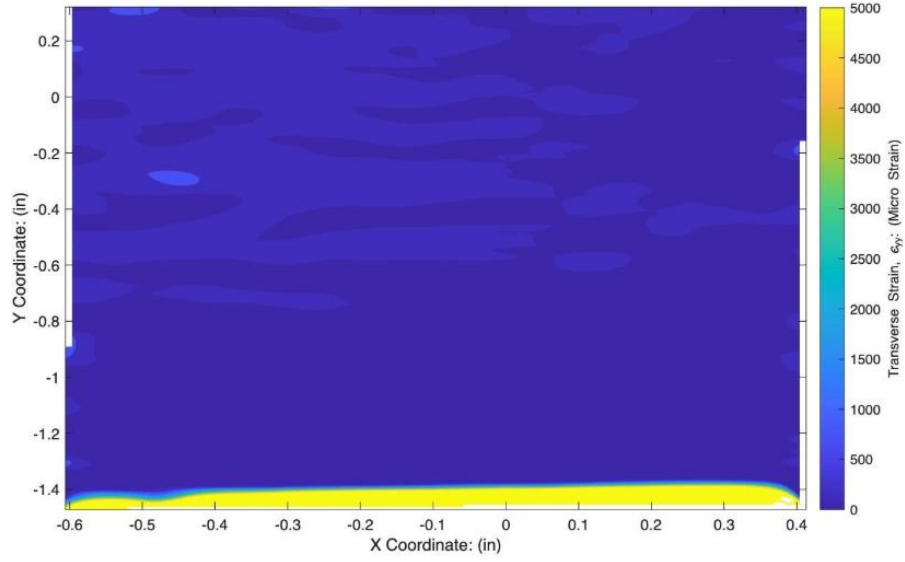


Figure 3.10: Tensile specimen Grid-2 ϵ_{yy} microstrain contour map.

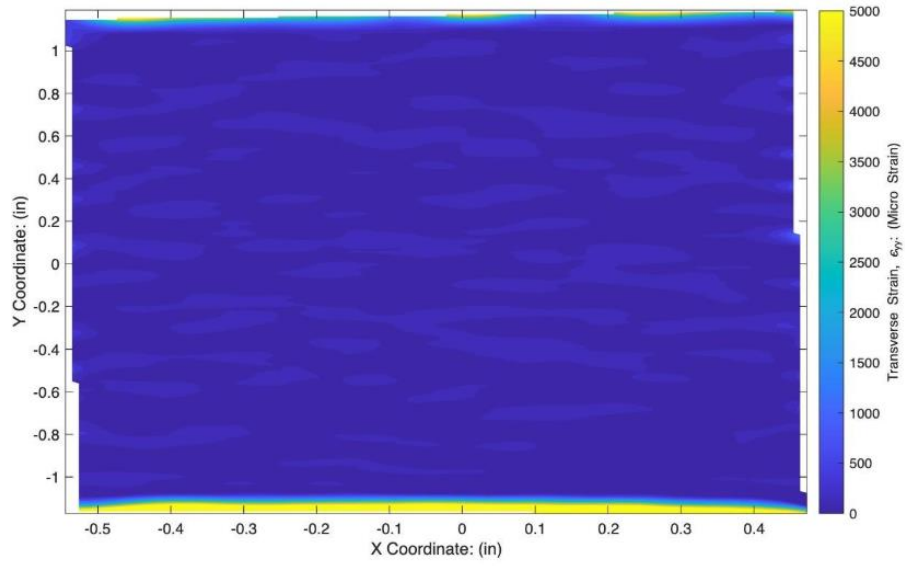


Figure 3.11: Tensile specimen Honeycomb-1 ε_{yy} microstrain contour map.

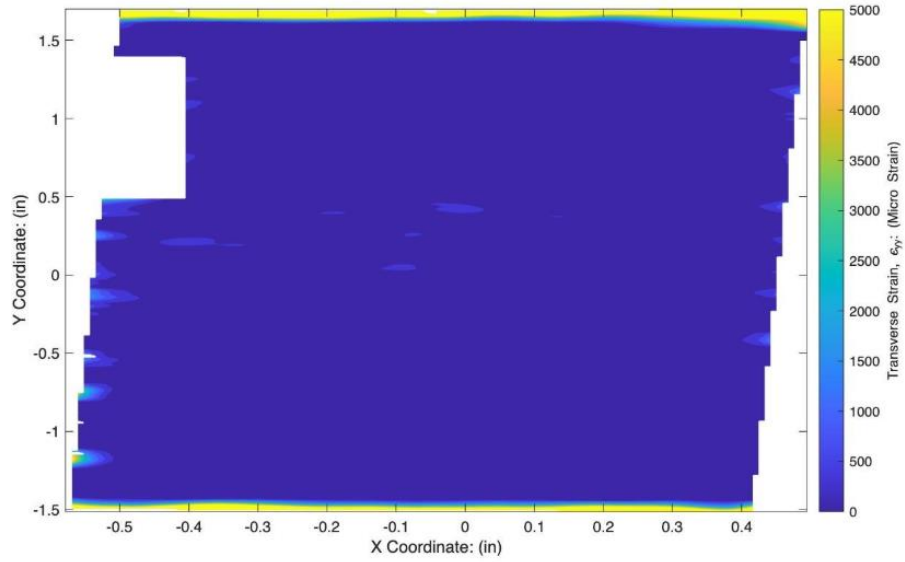


Figure 3.12: Tensile specimen Honeycomb-2 ε_{yy} microstrain contour map.

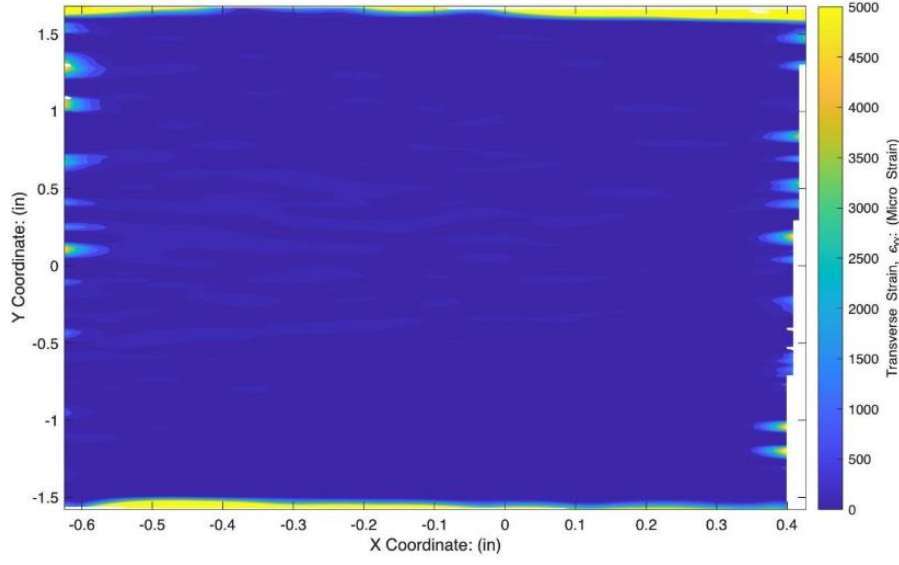


Figure 3.13: Tensile specimen Triangular-2 ε_{yy} microstrain contour map.

The contour maps have minimal variation, with exceptions around the edge where slight speckle pattern delamination likely occurred. This minimal variation is expected since the Instron applies a given displacement along the horizontal component equally. The data was reduced similarly to the bending data by selecting three lines of data at set horizontal values: one near the geometric center, one on the left-hand side, and one on the right-hand side. This provides a range of data in both directions to capture a macro-scale response. The data was reduced by taking an average of all data points to obtain $\varepsilon_{yy,avg}$. An applied force was recorded and corresponded to each DIC image capture. Both the ε_{yy} data and corresponding applied force were recorded in Table 3.3.

Table 3.3: Tensile Specimen Applied Force Terms

Specimen	Average ε_{yy}	Applied Force (N)	Area (m ²)	Stress (MPa)
Grid-1	5.9453×10^{-4}	218	1.644×10^{-4}	1.3261
Grid-2	7.141×10^{-4}	250	1.648×10^{-4}	1.5167
Honeycomb-1	2.4612×10^{-4}	48.3	1.648×10^{-4}	0.2930
Honeycomb-2	2.6824×10^{-4}	68	1.619×10^{-4}	0.4201
Triangular-2	2.0061×10^{-4}	87.02	1.680×10^{-4}	0.5178

The applied force is related to the normal stress using Equation 3.5.

$$\sigma_{norm} = \frac{F_{norm}}{A} \quad (3.5)$$

The A term is the area in square meters. This was calculated as the product of the thickness t and width w recorded in Table 2.2. The resulting area and normal stress values are shown in Table 3.3. This is then related to the tensile elastic modulus using Equation 3.6.

$$E_T = \frac{\sigma_{yy}}{\varepsilon_{yy}} \quad (3.6)$$

This can also be thought of as the slope of the linear elastic region of a common stress-strain curve. Characterizing a full stress-strain curve is difficult using DIC; however, the stress-to-strain ratio remains the same throughout the elastic region. Using Equation 3.6, the results for the tensile tests are presented in Table 3.4.

Table 3.4: Tensile Specimen Calculated Strength and Build Properties

Specimen	E_T (MPa)	Force at Fracture (N)	Print Time (mm:ss)
Grid-1	2.2306×10^3	1904	44:22
Grid-2	2.1239×10^3	678.3	44:22
Honeycomb-1	1.1906×10^3	1410	52:30
Honeycomb-2	1.5662×10^3	1935	52:30
Triangular-2	2.5812×10^3	2099	44:34

Additionally, the force at failure and print time are presented here. Similar to the prior case, the failure force was recorded right as visible fiber separation was observed, which frequently occurred in a rapid and distinct manner, making it easy to observe. The print time was calculated using Bambu Studio software on a Bambu Lab X1E 3D printer.

The final results show consistency with some metrics and inconsistent values with others. Both the bending and tensile test specimens have print times in the same relative order of grid, triangular, and honeycomb in ascending order. The compressive elastic modulus results had some difference across the same infill, but they were distinctly ranked relative to one another, with the honeycomb infill ranking the highest and the grid patterns the lowest. The tensile elastic modulus results were not similar, with the triangular pattern ranking the highest and the honeycomb the lowest. There was relative agreement between the grid pattern samples but not between the honeycomb samples. The applied moment at fracture and force at fracture results were inconsistent for both tests, with a notable difference of 1226 N between the two grid patterns. This may indicate the inherent uncertainty in 3D printed materials at fracture, particularly the impact of printed defects at the small scale examined here. Additionally, all tension tests failed along the interface between the clamping and test regions, with infills of 100% and 10%, which likely contributed to the large variations. DIC could not be used at large deflections because delamination impacts would be amplified,

though delamination was present in all tests, mainly on the edges of the test specimens.

Summary and Conclusions

The purpose of this investigation was to experimentally determine how the infill geometry, grid, honeycomb, or triangular, of 3D printed PLA specimens affects their mechanical performance under two distinct loading conditions: four-point bending and uniaxial tension. The investigation was executed using a controlled Instron load frame, with the resulting strain fields captured by Digital Image Correlation. Data reduction in MATLAB allowed for calculation of the compressive elastic modulus E_C from the bending tests and the tensile elastic modulus E_T from the tension tests, along with corresponding fracture loads.

The results successfully addressed the problem, demonstrating that infill geometry is a significant determinant of mechanical properties, though different infills excelled under different loading types. Under compression from bending, the honeycomb infill exhibited the highest compressive elastic modulus, followed by the triangular and then the grid patterns. Conversely, under pure tension, the triangular infill achieved the highest tensile elastic modulus, followed by grid, with honeycomb ranking the lowest. The overall findings confirm the initial hypothesis that infill design plays a critical role in stiffness, with the honeycomb pattern providing superior compressive resistance and triangular offering the best tensile stiffness among the tested geometries. We addressed the main crux of our research question, though we did not observe the full range of stress-strain behavior due to not being allocated the sensors to do so.

Error Analysis

We dealt with repetitive delamination of the DIC stickers throughout the bending testing. Despite the adhesive on the back of all DIC patterns, as well as additionally taping the edges of the stickers to our test specimens, we observed delamination in all cases at varying test subject displacements. This was not an issue for the tension tests. This delamination destroys the accuracy of the DIC system by generating wrinkles in the speckle pattern that distort the image processing. Thus, we had to discard most DIC data from high bending beam displacements. DIC inherently requires removing problematic areas of incongruent data from the area of interest, as can be seen in Figures 3.1 through 3.4, and the wrinkles massively exacerbated this problem to the point that there was no useful data to extract from high-displacement images with substantial speckle pattern delamination.

In the case of the tension specimens, we had to manufacture the ends of the dogbone at 100% infill in order for them not to be crushed by the Instron jaws. However, we manufactured the area of interest, the long, narrow section of the dogbone, at 10% density in order to observe more elastic behavior, require less force and displacement during testing, reduce material consumption, and reduce manufacturing time. Thus, the manufactured dogbones had a sharp change in infill density from 100% to 10% at the beginning of the neck section of the specimen, and this is where failure occurred every time during testing, as shown in Figure 5.1.



Figure 5.1: Fracture area of tensile test specimen.

This repetitive behavior indicates that this was a weak point in the tension dogbone and potentially reduced the ultimate strength of the specimen. In the future, a more gradual change in infill density should be employed, if possible, or this location of sudden infill density change should be placed at a less sensitive area.

All transducers and instrument recording devices have inherent uncertainty. However, in the scope of this project, the errors of the Instron machine and strain gauges are small compared to the DIC delamination of the bending beams and the fracture weak point of the tension dogbones.